Venkata Lakshmi Parimala Pasupuleti

Full Stack Software Engineer · PropTech & Apartment AI · React/TypeScript Frontend · API Design · Full-Stack Product Delivery · Startup Ownership

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PROFESSIONAL SUMMARY

Full-stack software engineer with 8 years building customer-facing products in apartment and affordable housing software - the exact domain EliseAI operates in. I build front-to-back: React/TypeScript frontends that translate complex workflows into clean user experiences, scalable REST APIs that power them, and the modular reusable components that hold it all together across a multi-tenant SaaS platform. At Entrata, I launched 20+ production modules used daily by property managers, leasing agents, and compliance teams at 100K+ housing units - working directly with product, operations, and clients across 120+ sprints without a playbook. AI on FHIR is the full-stack proof point: React.js/TypeScript frontend plus Python/Flask backend translating natural-language queries into FHIR API calls with HIPAA-compliant RBAC. I thrive in startup environments where you own the feature end-to-end, bias for action, and ship fast. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Full-Stack & API Engineering: TypeScript, React.js, JavaScript, Python (8yr), FastAPI, Flask, REST APIs, PostgreSQL, MySQL, Oracle, Redis, RabbitMQ, Docker, AWS (S3, EC2, RDS, Glue), CI/CD (GitHub Actions), Microservices, Multi-Tenant SaaS, Linux, Git

AI & Autonomous Systems: Claude Code, LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, Hallucination Detection, RAG Pipelines, ChromaDB, Embeddings, Prompt Engineering, Anthropic API, OpenAI API, MCP, HITL, Guardrails, OCR, spaCy, BERT, NLP, PyTorch, scikit-learn

Reliability & Delivery: Modular Reusable Components, Frontend User Experience, Cross-Functional Stakeholder Delivery, Testing Strategy (Unit, Integration, End-to-End), Observability & Monitoring, Performance Engineering, Production Debugging, Continuous Integration, PHI Compliance, Data Governance, Code Reviews & Mentoring, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - End-to-End Delivery

- Scoped and launched a production-grade agentic AI system (LangGraph, FastAPI, Python) from ambiguous client requirements to deployment - owned full-stack architecture, frontend integration, testing, observability, and stakeholder handoff with full autonomy; used Claude Code throughout to compress the build cycle and ship working software in weeks.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and system design.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **20+ modular, reusable production modules launched for apartment operators** - customer-facing features for property managers, leasing agents, and compliance teams at 100K+ housing units; owned the HUD income-limit pipeline for 7 years (federal data → 180,000 domain records powering eligibility decisions); shipped across 120+ Agile sprints working directly with product, compliance, and operations stakeholders - no micromanagement, full ownership from design to deployment.
- **Scalable APIs and full-stack infrastructure built to run reliably at scale** - REST API design across multi-tenant SaaS; RabbitMQ async processing with retry logic, idempotency guarantees, and dead-letter handling; Docker containerization (50% scalability improvement, 30% deployment time reduction); AWS S3/EC2/RDS/Glue; Redis caching; 200+ SQL query optimizations (30% performance gain); 99.5% uptime; 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents 7yr).
- **Delivered with startup urgency and cross-functional ownership** - GitHub Actions CI/CD; maintained sub-10 bug count for 6+ consecutive months; built internal debugging tools reducing triage time by 40%; 300+ peer code reviews; leveraged Claude Code and AI-assisted development to accelerate iteration; mentored 5+ engineers.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for apartment management and affordable housing compliance; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AI on FHIR: Full-Stack Customer-Facing NLP System - React/TypeScript + Python

Full-Stack · React/TypeScript · REST APIs · Customer-Facing

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack customer-facing AI application built front-to-back: React.js/TypeScript frontend with modular component architecture translating natural-language queries into FHIR REST API calls; Python/Flask backend with BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with authentication, authorization, and full data isolation; end-to-end ownership from user experience design to API design to deployment.

LeaseI: Agentic AI Platform for Apartment Compliance - Concept to Production

PropTech · Autonomous Systems · End-to-End

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Three-surface production-grade AI platform built solo from discovery through deployment - document intelligence (OCR, section-aware chunking, 100% citation correctness), 8-agent LangGraph support investigation (autonomous tool routing, memory, HITL escalation), and tenant self-service with deterministic guardrails. Full testing strategy, observability, and CI integration - 93.6% precision, 87.9% F1 across 1,240 apartment compliance test cases.

AgentProbe: LLM Evaluation with CI-Integrated Regression Gate

Observability · Testing · Open Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production LLM evaluation platform built end-to-end: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking regressions before they ship, multi-provider routing for cost optimization. Zero regressions shipped across 1,240+ test cases. Open-source.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017